Jian Zhou (Tony) Tu

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EDUCATION

University of Utah Salt Lake City, UT, USA
M.S., Computing (Artificial Intelligence track), GPA 4.0/4.0 Expected May 2027

- Thesis **(***working title***):** Task-Conditioned Information Gathering for Sequential Manipulation Under Occlusion. Advisor: Prof. Tucker Hermans.
- Relevant Coursework: Probabilistic Deep Learning (*in progress*); Motion Planning; Advanced AI; Deep Learning; Algorithmic Foundations of Robotics; Artificial Intelligence.

Carleton University Ottawa, ON, Canada
Bachelor of Computer Science Honours, Artificial Intelligence and Machine Learning Stream,
Co-operative Education, With High Distinction, GPA: 10.57/12.00 June 2025

- Thesis title: Autonomous Helicopter Landing Point Range Estimation Through Flight Trajectory and Landing Site Lidar Point Cloud Using a Supervised Machine Learning Approach. Advisor: Prof. Matthew Holden.

RESEARCH INTERESTS

Robotic manipulation under uncertainty, with interests in:

- Active perception and task-directed information gathering
- Probabilistic scene representations connecting perception and motion planning
- Learning-based planning and decision-making under partial observability, with safety guarantees

RESEARCH EXPERIENCE

University of Utah, Learning Lab for Manipulation Autonomy (LL4MA Lab) Sept 2025 – present
Graduate Research Assistant (since Jan 2026); Advisor: Prof. Tucker Hermans

Information gathering for manipulation under occlusion (M.S. thesis) Jul 2026 – present

- Developing a method for a fixed-camera robot arm to retrieve a target from a scene of unknown objects. The robot first plans an optimistic retrieval trajectory that treats unobserved space and other objects as free while still avoiding the static environment, and uses its swept volume as the task-relevant region. Unobserved space in that region is resolved by peeking (lifting an occluder, observing behind it, and returning it), and objects blocking it are relocated; the robot executes only once the region is observed and clear.
- Built the full system: a voxel occupancy belief with entropy-based uncertainty, and a decision layer that executes the retrieval once the swept volume is fully observed and free, and otherwise selects whichever of peeking or relocating maximizes expected information gain over that region, using cuRobo motion plans.
- Evaluated in kinematic simulation on the 20 scenes of the SceneReplica benchmark (one designated target per scene), retrieving the target in every scene, typically with at most one extra action. Demonstrated on a real KUKA iiwa in two comparable tabletop scenes; currently extending to shelf scenes.

Real-robot pick-and-place system (KUKA iiwa 7-DOF arm, Robotiq 2F-140 gripper) Sept 2025 – Aug 2026

- Worked with postdoc Dr. Martin Matak on known-object perception: implemented ICP-based localization, then proposed and built a custom YOLO detector (labeling and training) to provide initial masks for FoundationPose,

later replaced by color masking on single-color 3D-printed Objaverse objects; ran real-robot experiments together (Sept – Dec 2025)

- Sole developer since Jan 2026: extended Dr. Matak's cuRobo-based planning wrapper into a full pick-and-place pipeline, adding pre-execution trajectory preview with abort (Viser), joint pick-and-place planning before execution (Devendran Shanthi & Hermans, RA-L 2024), and grasp and placement re-scoring from the configurations the arm actually reaches (First Plan Then Evaluate, Matak et al.).
- Removed the one-object-per-color restriction by replacing color masking with perception for arbitrary objects: Fast-FoundationStereo depth, SAM2 segmentation fused with 3D clustering, and GraspGen grasps on partial point clouds; migrated the system from cuRobo v1 to v2 with a full codebase rewrite (May – Jun 2026).
- Explored partially reusing motion plans across scene changes to reduce replanning, using Prof. Hermans's certificate-based approach (OASIS): states and trajectory segments are certified individually, so still-valid portions of a plan are kept and only invalidated segments are replanned. Implemented the certificate backend (state and edge certificates, KD-tree storage, inverted index) in Isaac Sim and compared five planning variants on scenes of 3 to 16 objects. Certificates were never falsely safe and were largely retained within episodes, but cross-episode reuse was limited, and a fresh Lazy-PRM baseline remained 2 to 5× faster (early summer 2026).
- Redesigned hand-eye calibration: designed and 3D-printed a gripper-mounted ChArUco fixture with enlarged squares and markers for the camera's working distance, and an automated, replayable sequence of pre-recorded calibration poses (Jul – Aug 2026).

National Research Council Canada, Flight Research Laboratory, Ottawa May – Jul 2024

Research Assistant (co-op), Canadian Vertical Lift Autonomy Demonstration (CVLAD): Helicopter Landing-Phase Identification Project

- Sole developer of the project's landing-detection models: trained and compared supervised models on 6-DOF helicopter flight time-series data (MLP, CNN–MLP, RNN–MLP and CNN–RNN–MLP architectures in TensorFlow/Keras), evaluating different input feature sets.
- Developed a MATLAB app to streamline data preprocessing and manual labeling.
- Co-presented the results at the CASI ASTRO 2024 Conference and wrote the machine learning sections of the lab's internal technical report.

Carleton University, Advisor: Prof. Dave Campbell May – Aug 2021

Dean's Summer Research Intern - Undergraduate research student

- Eagle-nest detection from video: updated legacy TensorFlow/YOLO code, set up frame preprocessing, and exposed the network's pre-activation logits to support explainability of the model.

PROFESSIONAL EXPERIENCE

National Research Council Canada, Flight Research Laboratory, Ottawa May – Jul 2025

Software Developer (Contractor), Canadian Vertical Lift Autonomy Demonstration

- Developed TEMPS-2, a browser-based helicopter mission-planning tool compatible with QGroundControl, and wrote its user and developer handover guides.
- Features include an editable waypoint table with derived flight figures, an altitude-versus-terrain profile that flags legs below terrain, Canadian airspace layers in 2D and 3D, reusable mission segments, and export to .plan, .xlsx and .kml.

National Research Council Canada, Flight Research Laboratory, Ottawa Sept 2022 – Apr 2025

Software Developer (Co-op, and part-time), Canadian Vertical Lift Autonomy Demonstration

- Extended QGroundControl, open-source flight-planning software, for helicopter missions: airspeed or ground

speed per waypoint, altitude in AGL or AMSL, and automatic waypoint altitude updates via the Google Elevation API.

- Built TEMPS, a browser-based mission planner (JavaScript, Mapbox GL JS) that replicated QGroundControl's core features to replace it for mission planning, and helped integrate it into the lab's existing system.
- Prototyped a 3D pilot display in Unreal Engine 5 with Cesium, demonstrated through flight replay, and built a MATLAB tool that converts public surface and terrain elevation data (DSM, DTM) into layered 3D terrain models by extruding and stacking elevation contours.

Shared Services Canada, Ottawa May – Aug 2022

Data management analyst (co-op)

- Performed ETL and data migration (ECD to ONYX), wrote Excel VBA scripts to automate workflows, and maintained the configuration-management database.

TEACHING EXPERIENCE

Carleton University, Ottawa, ON, Canada Sept 2021 – Apr 2022

Teaching Assistant, Calculus (MATH 1007, MATH 1009)

- Held office hours, recorded video solutions, and graded tests and assignments with same-day turnaround.

PRESENTATIONS

"Development of Space Vehicle Landing Behavior for Planetary Surface Exploration Through Machine Learning and Emulation," CASI ASTRO 2024 Conference, Toronto, Canada, August 2024. Co-presented with H. Morry; presented the machine learning results.

LEADERSHIP AND SERVICE

FIRST Robotics Competition (FRC) teams, Canada and USA Sept 2019 – present

Mentor, Team 5933, Judge Memorial Catholic High School, Salt Lake City (Sept 2025 – present)
Lead Mentor, Team 10464, Blyth Academy Glebe, Ottawa (Sept 2024 – Apr 2025)
Lead Mentor, Team 8729 Sparkling H2O, Spark Youth Robotics Club, Ottawa (Sept 2020 – Apr 2024)
Mentor, Team 7710 Mowat Mustangs, Sir Oliver Mowat Collegiate Institute, Toronto (Sept 2019 – Apr 2020)

- Taught programming, CAD and robot design to high-school students (20 - 50 students per team), and guided troubleshooting through each build season.
- As lead mentor, organized meetings and team roles and managed budget, fundraising and procurement.
- Team 8729 won its division at the FIRST Ontario Provincial Championship in 2023 and 2024, was a 2023 provincial finalist, and competed at the 2023 FIRST Championship (Houston, TX).

AWARDS AND HONOURS

Carleton University: A. Davidson Dunton Scholarship (2024); Michael Oliver Scholarship (2021)

SKILLS

- **Programming:** Python, C++, MATLAB, JavaScript/TypeScript
- **Robotics and ML:** PyTorch, TensorFlow/Keras, ROS 2, cuRobo; real-robot experience with the KUKA iiwa and Robotiq 2F-140; hand-eye calibration; CAD and 3D printing
- **Tools:** Git, Docker, Linux, Unreal Engine 5, Unity